STAFFORD HO SHENG XIAN

8506 0453 ·stafford_ho@rocketmail.com ·linkedin.com/in/stafford-ho ·staffordho.github.io

Summary

Final-year Robotics Systems Engineering undergraduate with hands-on experience in research, industrial automation, and interdisciplinary collaboration. Experienced in developing robotic proof-of-concepts (PoCs), integrating soft robotics with AR/VR-led systems, and deploying IoT-enabled control systems with computer vision for industrial and research applications. Strong background in C/C++, Python, structured text (PLC), and ROS, complemented by leadership roles in academic committees and national-level collaborations. Passionate about combining systems-level thinking with practical prototyping for impactful engineering solutions.

Work Experience

Intern – Robotics Solutions, OMRON *Jan 2025–Dec 2025*

- Deployed a Mobile-Manipulator (MoMa) system in collaboration with stakeholders, automating a 24/7 labour-intensive process, enabling operator upskilling and redeployment to higher-value tasks, while improving overall throughput by 10%.
- Co-authored decision-making & navigation stack for MoMa and mapped out a real-world environment for traversal and docking.
- Programmed Pick-and-Place routines for MoMa at defined pickup/dropoff points, including lab and site integration with error-recovery logic.
- Created Digital Twin simulation scenarios in Mobile Planner software using Virtual Fleet Manager (VFM) for path-planning, traffic management and in-lab validation prior to site deployment.
- Built Pick-and-Place routines in TMflow software on OMRON TM Robotic arm and delivered stakeholder Proof-Of-Concept (PoC) demonstration videos for picking circular, hollow-centre objects.
- Wrote a script for an automated 2D imaging pipeline of OMRON SENTECH Liquid Lens Industrial Cameras using OpenCV for image processing and StApiPy for device control.
- Developed a generic CPP Inference Application Programming Interface (API) for Open Neural Network Exchange (ONNX) formats integrated with OMRON vision controllers built on ONNX Runtime (ORT) for optimization of inference speed up to 10%.
- Performed a Computer Vision application demonstrating the deployment of several object detection models executed in series on the STM32 Discovery kit.
- Conducting senior thesis on the optimization of inference engines with machine learning on OMRON vision controllers, and studying the speed-accuracy trade-offs to improve industrial vision performance.

Student Assistant – Research, Singapore Institute of Technology *Sep 2024–Sep 2025*

- Conducted data transmission experiments and testing related to edge intelligence and cloud connectivity via multiple physical interfaces.
- Contributed to a MINDEF-funded project developing an online process control system that automatically dispenses biocide to prevent fuel-tank biocontamination. The work culminated in a live demonstration and technology transfer to MINDEF directorates on 30 Aug 2025, and a journal paper is now under peer review.
- Currently advising on the application and integration strategy of soft robotics, assisting an Infocomm Technology (ICT) faculty-led student team contributing to a paper on adaptive visual-language navigation. Applications for NVIDIA academic grant and poster submission for NVIDIA GTC 2026 are underway.

Student Assistant – Academic , Singapore Institute of Technology *Jan 2025–Feb 2025*

- Designed and wrote technical documentation used for academic workshops under the specialization module MEC3272: Mechatronics Design (BEng in Mechanical Engineering).
- Diagnosed and resolved issues with STM32 Nucleo boards for academic labs under modules RSE2701: Analog and Digital Electronics & RSE3702: Embedded Systems (BEng in Robotics Systems).

Student Assistant – Projects, Singapore Institute of Technology *Aug 2023–Sep 2023*

- Led a 3-member team to fabricate a PoC Net Launcher for the National Environment Agency (NEA), achieving implementation within 4 weeks.
- Created conceptual 3D sketches and CAD drawings during sprints and streamlined project workflows & deliverables in an electronic logbook.

Intern – Technology Development, Systems on Silicon Manufacturing Company *Mar 2019–Aug 2019*

- Delivered presentations at department meetings using agile methodology.
- Developed a digital QR code system using Excel Macros, increasing task efficiency by 30%.

Intern – Chief Controls Office, Barclays *Jun 2018, Sep 2018–Oct 2018*

- Completed 2 weeks of professional training focused on optimizing soft skills.
- Developed a central digital data repository for file migration, improving efficiency by 40%.

Education

Bachelor of Engineering (BEng) with Honours in Robotics Systems, Singapore Institute of Technology
Aug 2022–Est. May 2026

- President, Student Management Committee (AY2022/23)
- Year 2 Representative, Student Management Committee (AY2023/24)
- Year 3 Representative, Student Management Committee (AY2024/25)
- Year 4 Representative, Student Management Committee (AY2025/26)
- Student Coach, Catalyst-SIT@Dover/ ProjectHub-SIT@Punggol (AY2024/25, AY2025/26)
- Ngee Ann Kongsi Scholarship Recipient (AY2025/26)

Diploma in Electronic Systems, Nanyang Polytechnic *Apr 2017–Apr 2020*

Technical Skills

- **Software**: Android Studio, Capella, Isaac Sim, Mobile Planner, MQTT Explorer, MQTTX, RoboDK, SOLIDWORKS, Sysmac Studio, TMflow, Visual Components
- **Programming Languages**: C, C++, Kotlin, Ladder logic, Python, Structured Text
- **Embedded Systems**: ArduinoIDE, Raspberry Pi, STM32CubeIDE, STM32CubeMX, STM32CubeProg
- **Frameworks**: Robot Operating System (ROS, ROS2), micro-ROS